

Ammonia–N in Drinking and Surface Waters, Domestic and Industrial Wastes

1 SCOPE

This method covers the determination of Ammonia–N in seawater.

2 RANGE OF APPLICATION

Range	0.02 – 1.0 mg N/L (1.4 – 71 µM)
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3 METHOD PERFORMANCE SUMMARY

By EPA Procedure	MDL = 0.006 mg N/L (0.429 µM)
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4 METHOD PRINCIPLE

At alkaline pH, ammonia in the sample reacts with hypochlorite (HClO^-), as previously liberated from dichloroisocyanurate. The chloramine formed then reacts with salicylate, at pH at least 12.8, in presence of nitroferricyanide. During static incubation at 40 °C, a blue-green indophenol dye forms, which is measured photometrically at 660 nm.

5 REFERENCES

Methods for the Determination of Inorganic Substances in Environmental Samples, EPA 600/R 93/100, August 1993: Method 350.1, Revision 2.0.

6 INTERFERENCES

Dissolved calcium and magnesium may precipitate under the alkaline reaction conditions to cause positive bias. Heavy-metal cations may precipitate similarly. The EDTA and citrate in the buffer help to counteract this effect.

If samples have been preserved using sulfuric acid, then samples must be neutralized prior to analysis.

When required, interferences are removed by manual distillation of samples. Refer to EPA 350.1 Rev. 2.0 for the distillation procedure and for discussion of potential interferences during distillation.

Bias from sample turbidity or color is correctable, using sample blanking feature of AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the safety data sheets (SDS) for details on safe handling of chemicals.

<u>Chemical</u>	<u>Formula</u>	<u>C.A.S. No.</u>	<u>Safety Classification</u>
Ammonium chloride, anhydrous	NH ₄ Cl	12125-02-9	GHS07
EDTA	C ₁₀ H ₁₆ N ₂ O ₈	60-00-4	GHS07
Sodium dichloroisocyanurate, anhydrous or dihydrate	C ₃ Cl ₂ N ₃ NaO ₃	2893-78-9	GHS03, GHS07, GHS09
	C ₃ Cl ₂ N ₃ NaO ₃ •2H ₂ O	51580-86-0	GHS07, GHS09
Sodium chloride*	NaCl	7647-14-5	--
Sodium hydroxide	NaOH	1310-73-2	GHS05
Sodium nitroferrocyanide dihydrate	Na ₂ [Fe(CN) ₅ NO]•2H ₂ O	13755-38-9	GHS06
Sodium salicylate	C ₇ H ₅ NaO ₃	54-21-7	GHS07
Sodium citrate dihydrate	C ₆ H ₉ Na ₃ O ₉	6132-04-3	--
Sodium potassium tartrate tetrahydrate	C ₄ H ₄ KNaO ₆ •4H ₂ O	6381-59-5	--
Sulfuric acid	H ₂ SO ₄	7664-93-9	GHS05
GHS01: Danger – Explosive GHS02: Danger – Flammable GHS03: Danger – Oxidizing GHS04: Warning – Compressed Gas GHS05: Warning/Danger – Corrosive GHS06: Danger – Toxic GHS07: Warning – Irritant GHS08: Danger – Health Hazard GHS09: Warning/Danger – Aquatic Toxicity			

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

*For use when making synthetic seawater only

8 REAGENT SOLUTIONS

Use reagent-grade chemicals, certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination; therefore Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 SULFURIC ACID, 5 normal

Sulfuric acid, concentrated

70 mL

DI Water

dilute to 500 mL

Slowly add 70 mL concentrated sulfuric acid to about 400 mL of deionized water. **CAUTION**, the container will become very warm. Cool to room temperature and dilute to 500 mL with deionized water.

Store at room temperature. This solution is available commercially. Stable for 6 months.

8.2 SODIUM POTASSIUM TARTRATE, 20% w/v (pH 7-8)

Sodium potassium tartrate tetrahydrate	200 g
Sodium hydroxide	about 5 pellets
Sulfuric acid, 5 normal (above)	pH adjustment
DI water	dilute to 1 L

Dissolve 200 g sodium potassium tartrate tetrahydrate in about 400 mL ammonia-free deionized water. Add 5 pellets of sodium hydroxide. Boil the solution, with stirring, for 1 hour to drive off ammonia contamination. Add make-up water as needed. Cover the flask and cool to room temperature. Add acid dropwise (sulfuric or hydrochloric, about 5 normal) to reach pH of 7.5 ± 0.5 . Dilute to 1 L.

Store at 4°C. Stable for 6 months.

8.3 BUFFER

EDTA	5.2 g
Sodium hydroxide	3.0 g
Sodium citrate dihydrate	7.5 g
Sodium potassium tartrate, 20% w/v, above	dilute to 100 mL

To a 100 mL beaker, add approximately 50 mL of sodium potassium tartrate, 20% w/v. Add 5.2 g EDTA and initially 2 g sodium hydroxide. Heat slightly to dissolve. Then transfer to a volumetric flask and add 7.5 g of sodium citrate dihydrate and the final 1.0 g of sodium hydroxide. Stir to dissolve – adding additional sodium potassium tartrate solution as needed. Once dissolved and cooled to room temperature, dilute to 100 mL using the 20% sodium potassium tartrate solution.

Store at 4°C. Stable for 6 months.

8.4 SALICYLATE

Sodium salicylate	65 g
Sodium citrate dihydrate	65 g
Sodium nitroferrocyanide dihydrate	0.5 g
DI water	dilute to 500 mL

Add 65 g sodium salicylate, 65 g sodium citrate dihydrate and 0.5 g sodium nitroferrocyanide dihydrate to a 500 mL volumetric flask, containing approximately 400 mL deionized water. Stir to dissolve, then dilute to 500 mL. Filter this reagent, e.g., using filter paper.

Store at 4°C. Stable for 2 months.

8.5 DICHLOROISOCYANURATE (DCI)

Sodium hydroxide	6.0 g
Sodium dichloroisocyanurate, anhydrous	0.330 g
or	
Sodium dichloroisocyanurate dihydrate	0.383 g
DI Water	dilute to 100 mL

Add 6.0 g sodium hydroxide pellets to approximately 50 mL deionized water in a 100 mL volumetric flask. Dissolve completely, then cool to near room temperature. Add 0.330 g sodium dichloroisocyanurate, anhydrous or 0.383 g sodium dichloroisocyanurate dihydrate and stir to dissolve. Dilute to 100 mL with deionized water.

Store at 4°C. Prepare fresh before use, at least every other day.

Note: It is recommended to store the dry chemical at 4°C.

9 STANDARD SOLUTIONS

9.1 AMMONIA STOCK STANDARD SOLUTION (1000 mg N/L)

Ammonium chloride, anhydrous 3.819 g
DI water dilute to 1 L

Dissolve 3.819 g ammonium chloride, anhydrous (dried at 105°C) in about 800 mL deionized water in a 1 L volumetric flask. Dilute to the mark with deionized water.

Store at 4°C. Stable for 1 month.

9.2 CALIBRATION DILUENT (SYNTHETIC SEAWATER)

Sodium chloride 35 g
DI water dilute to 1 L

Dissolve 35 g of sodium chloride in about 800 mL of deionized water in a 1 L plastic volumetric flask. Dilute to 1 L with deionized water. Store in a plastic container.

***Note:** The concentration of sodium chloride can be adjusted to match the salinity of the seawater samples.

9.3 AMMONIA INTERMEDIATE STANDARD SOLUTION (50 mg N/L)

AMMONIA Stock Standard (1000 mg N/L, above) 10 mL
Calibration diluent dilute to 200 mL

Transfer 10 mL AMMONIA Stock Standard (1000 mg N/L) to a 200 mL volumetric flask. Fill to the mark with calibration diluent and invert to mix.

Store at 4°C. Stable for 1 week.

9.4 AUTO CALIBRATION – Ammonia 1.0 mg N/L solution

AMMONIA Intermediate Standard (50 mg N/L, above) 5 mL
Calibration diluent dilute to 250 mL

Transfer 5 mL AMMONIA Intermediate Standard (50 mg N/L) to a 250 mL volumetric flask. Fill to the mark with calibration diluent and invert to mix.

Store at 4°C. Stable for 1 week.

9.5 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, mg N/L	0	0.02	0.05	0.10	0.25	0.50	0.75	1.00
Volume (mL) Ammonia Auto Calibration Standard diluted as indicated with calibration diluent	--	2.0	5.0	10	25	50	75	100
Final Volume, mL	100	100	100	100	100	100	100	100

Store at 4°C. Stable for 1 week.

10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected in plastic or glass containers.

Samples should be analyzed as soon as practical after collection.

Samples can be frozen as long as thawed completely directly before analysis.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method document are only suggestions.

Any reagents having visible particulates should be filtered.

Pour calibration standards, quality control solutions, and samples into sample cups and reagent wedges, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent vessels and a waste collection bottle in place.

11.1 CHEMISTRY TROUBLESHOOTING TIPS

The concentration of active chlorine in the DCI reagent is crucial for obtaining full color development and low values for method detection limit.

When first evaluating method performance or if color development and linearity of the calibration line become unsatisfactory, check reaction pH. Measure the pH, after dispensing scaled-up test volumes of samples and reagents into a beaker. The optimum pH range of the mixture is 12.8 to 13.1. The table below shows the test volumes as scaled up 50-fold.

REAGENT	Test vol., μL	pH test vol., mL
Sample matrix (as preserved)	300	15
Reagent 1 (Buffer)	100	5.0
Reagent 2 (Salicylate)	100	5.0
Reagent 3 (DCI)	50	2.5

Measured pH of the ammonia reaction = _____

If the seawater samples being run do not contain high levels of magnesium or calcium ions, then the buffer can be diluted in half. To check if this acceptable, perform the beaker test described above with a typical sample. Gently heat to check for precipitation. If no precipitate, then the diluted buffer is acceptable to use.

It is recommended to rinse out sample cups with the sample prior to use to avoid contamination.

With software version 2.4.5 or older, it is recommended to run a manual calibration.

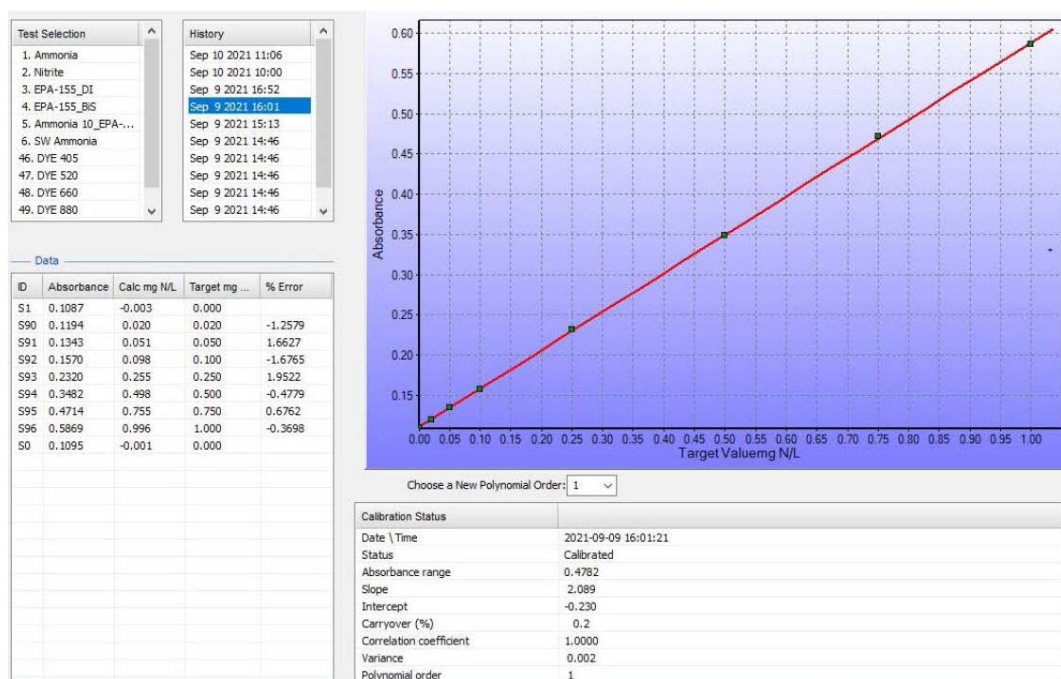
12 TEST PARAMETERS

PARAMETER	AQ SETTING
Test name	Ammonia_SW
Units	mg N/L
Decimals	3
Test type	End Point
Sample volume (µL)	300
Water volume (µL)	0
Number of mixes	2
Cuvette primes	2
Cuvette washes	2
Baseline on Wash	Ticked
Reaction time (seconds)	600
Wavelength (nm)	660
Polynomial order	1
Number of reagents	3
1. Buffer (µL)	100
2. Salicylate (µL)	100
3. DCI (µL)	50
Advanced Test Parameters	☒ Eliminate Air from Test Transfer ☒ Extra Debubbling Action

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.11 AU

Sensitivity Range (1.0 mg N/L): 0.46 – 0.70 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover: Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A - B)}{C} \times 100\%$$

Where,

A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = calibration absorbance range

14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.02 mg N/L (1.429 µM) ammonia standard in artificial seawater***

Concentration (mg N/L)

	0.025	0.019	0.022	0.026
	0.017	0.022	0.024	0.022
	0.018	0.018	0.026	0.022
	0.015	0.020	0.025	0.021
	0.021	0.016	0.026	0.020
	0.020	0.018	0.026	0.020
	0.017	0.021	0.023	0.021
	0.018	0.021	0.020	0.019
	0.018	0.019	0.023	0.020
	0.019	0.024	0.021	0.020
	0.018	0.016	0.029	0.023
	0.018	0.020	0.026	0.021
	0.014	0.018	0.023	0.021
	0.019	0.017	0.026	0.018
	0.017	0.023	0.025	0.023
Statistics				
Mean	0.018	0.019	0.024	0.021
Std. Deviation	0.003	0.002	0.002	0.002
MDL(mg N/L)*	0.007	0.006	0.006	0.005
MDL (µM)	0.478	0.453	0.441	0.360

14.2 PERFORMANCE STUDY

Replicates of 0.05 mg N/L (3.571 µM) ammonia standard in artificial seawater)***

Concentration (mg N/L)

	0.048	0.042	0.045	0.053
	0.049	0.044	0.046	0.054
	0.045	0.044	0.042	0.058
	0.054	0.047	0.045	0.055
	0.053	0.044	0.039	0.055
	0.046	0.044	0.044	0.056
	0.048	0.047	0.041	0.058
	0.049	0.043	0.045	0.052
	0.051	0.046	0.042	0.058
	0.053	0.042	0.043	0.050
Statistics				
Mean	0.050	0.044	0.043	0.055
Std. Deviation	0.003	0.002	0.002	0.003
% RSD**	6.2	4.1	5.1	5.0
% Recovery	99.2	88.6	86.4	109.8

14.3 PRECISION STUDY

Replicates of 0.5 mg N/L (35.714 µM) ammonia standard in artificial seawater***

Concentration (mg N/L)

Statistics				
Mean	0.515	0.521	0.519	0.490
% Recovery	103.1	104.3	103.8	98.1
Std. Deviation	0.009	0.020	0.006	0.003
% RSD**	1.7	3.7	1.2	0.7
Maximum	0.537	0.540	0.528	0.495
Minimum	0.508	0.469	0.505	0.484

Replicates of 1.0 mg N/L (71.429) ammonia standard in artificial seawater***

Concentration (mg N/L)

Statistics				
Mean	0.970	0.987	1.018	1.007
% Recovery	97.0	98.7	101.8	100.7
Std. Deviation	0.013	0.010	0.009	0.007
% RSD**	1.3	1.1	0.9	0.7
Maximum	0.991	1.000	1.030	1.019
Minimum	0.942	0.971	1.007	0.993

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9.

15 REVISIONS

September 2021	Rev 0	AQ400 method created and data collected
May 2022	Rev 1	Minor editorial changes Added sulfuric acid to reagent list